

NAME(print): _____

Note: There is a total of 49 points available in this test. The test is graded out of 37 points and any number of points (up to 12) exceeding this value will count as extra credit for Midterm 1.

1. (14 points) Answer each of the following questions precisely but briefly (no need of a proof).

(a) (2 points) Give the definition of an isomorphism of G -sets.

Solution: Let X and Y be G -sets. An *isomorphism of G -sets* is a bijection $\varphi : X \rightarrow Y$ such that $\varphi(g \cdot x) = g \cdot \varphi(x)$ for all $g \in G$ and $x \in X$.

(b) (2 points) Give the definition of a transitive G -set.

Solution: A G -set X is called *transitive* if for every pair of elements $x, y \in X$, there exists some $g \in G$ such that $g \cdot x = y$. Equivalently, X has exactly one orbit under the action of G .

(c) (2 points) Give a representative for each of the **distinct** irreducible permutation representations of A_4 . How many of these are there in total?

Solution: The distinct irreducible permutation representations of A_4 correspond to the conjugacy classes of subgroups of A_4 . The subgroups of A_4 are:

- $\{e\}$ (trivial subgroup), giving the regular representation on 12 elements.
- $\langle (12)(34) \rangle \cong \mathbb{Z}_2$, giving a transitive representation on $[A_4 : \mathbb{Z}_2] = 6$ elements. (The three subgroups of order 2 are all conjugate in A_4 .)
- $\langle (123) \rangle \cong \mathbb{Z}_3$, giving a transitive representation on $[A_4 : \mathbb{Z}_3] = 4$ elements. (The four subgroups of order 3 are all conjugate in A_4 .)
- $V_4 = \{e, (12)(34), (13)(24), (14)(23)\}$, giving a transitive representation on $[A_4 : V_4] = 3$ elements.
- A_4 itself, giving the trivial representation on 1 element.

There are **5** distinct irreducible permutation representations in total.

- (d) (2 points) Give the definition of a Sylow p -subgroup of a finite group G .

Solution: Let G be a finite group and let p be a prime. Write $|G| = p^a m$ where $a \geq 1$ and $\gcd(p, m) = 1$. A *Sylow p -subgroup* of G is a subgroup $P \leq G$ of order p^a , i.e., a p -subgroup of G of the largest possible order.

- (e) (2 points) Give the definition of a linear representation of a group G .

Solution: A *linear representation* of a group G is a group homomorphism $\rho : G \rightarrow GL(V)$, where V is a vector space over a field k and $GL(V)$ denotes the group of invertible linear transformations from V to itself.

- (f) (2 points) What are the possible abelian groups of order 36 up to isomorphism?

Solution: By the Fundamental Theorem of Finite Abelian Groups, we factor $36 = 2^2 \cdot 3^2$ and list all possible abelian groups as products of cyclic groups corresponding to partitions of each prime-power factor:

- $\mathbb{Z}_4 \times \mathbb{Z}_9 \cong \mathbb{Z}_{36}$
- $\mathbb{Z}_4 \times \mathbb{Z}_3 \times \mathbb{Z}_3$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_9$
- $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3$

There are 4 abelian groups of order 36 up to isomorphism.

- (g) (2 points) Give a composition series for $\mathbb{Z}_{12}$. What are the composition factors of this series?

Solution: A composition series for $\mathbb{Z}_{12}$ is:

$$\{0\} \trianglelefteq \langle 6 \rangle \trianglelefteq \langle 3 \rangle \trianglelefteq \langle 1 \rangle = \mathbb{Z}_{12},$$

i.e., $\{0\} \trianglelefteq \mathbb{Z}_2 \trianglelefteq \mathbb{Z}_4 \trianglelefteq \mathbb{Z}_{12}$, where we identify $\langle 6 \rangle \cong \mathbb{Z}_2$, $\langle 3 \rangle \cong \mathbb{Z}_4$. The composition factors (successive quotients) are:

- $\langle 6 \rangle / \{0\} \cong \mathbb{Z}_2$,
- $\langle 3 \rangle / \langle 6 \rangle \cong \mathbb{Z}_2$,
- $\mathbb{Z}_{12} / \langle 3 \rangle \cong \mathbb{Z}_3$.

So the composition factors are $\mathbb{Z}_2, \mathbb{Z}_2, \mathbb{Z}_3$.

2. (5 points) If X is a G -set, prove that $G_{g \cdot x} = gG_xg^{-1}$ for any $g \in G$ and $x \in X$.

Solution: Recall that for any $y \in X$, the stabilizer is $G_y = \{h \in G : h \cdot y = y\}$.

($\subseteq$) Let $h \in G_{g \cdot x}$, so $h \cdot (g \cdot x) = g \cdot x$. Applying g^{-1} on the left, we get $g^{-1}hg \cdot x = x$, which means $g^{-1}hg \in G_x$. Therefore $h = g(g^{-1}hg)g^{-1} \in gG_xg^{-1}$.

($\supseteq$) Let $h \in gG_xg^{-1}$, so $h = gag^{-1}$ for some $a \in G_x$ (meaning $a \cdot x = x$). Then

$$h \cdot (g \cdot x) = (gag^{-1}) \cdot (g \cdot x) = g \cdot (a \cdot (g^{-1}g \cdot x)) = g \cdot (a \cdot x) = g \cdot x.$$

Hence $h \in G_{g \cdot x}$.

We conclude that $G_{g \cdot x} = gG_xg^{-1}$.

3. (5 points) Prove that if X is a transitive G -set, then X is isomorphic to the G -set G/G_x of left cosets of G_x in G , where $x \in X$ is any element.

Solution: Fix $x \in X$ and define $\varphi : G/G_x \rightarrow X$ by $\varphi(gG_x) = g \cdot x$.

Well-defined: If $gG_x = hG_x$, then $h^{-1}g \in G_x$, so $h^{-1}g \cdot x = x$, which gives $g \cdot x = h \cdot x$. Hence φ is well-defined.

G-equivariant: For any $g, h \in G$, we have $\varphi(g \cdot hG_x) = \varphi((gh)G_x) = (gh) \cdot x = g \cdot (h \cdot x) = g \cdot \varphi(hG_x)$.

Surjective: Let $y \in X$. Since X is transitive, there exists $g \in G$ with $g \cdot x = y$. Then $\varphi(gG_x) = y$.

Injective: If $\varphi(gG_x) = \varphi(hG_x)$, then $g \cdot x = h \cdot x$, so $h^{-1}g \cdot x = x$, meaning $h^{-1}g \in G_x$, and thus $gG_x = hG_x$.

Therefore φ is a G -set isomorphism, and $X \cong G/G_x$.

4. (5 points) Let H be a normal subgroup of a finite group G and assume that $|H| = p$, where p is a prime. Show that H is contained in every p -Sylow subgroup of G .

Solution: Since $|H| = p$, the subgroup H is a p -subgroup of G . By Sylow's theorem, every p -subgroup of G is contained in some Sylow p -subgroup, so there exists a Sylow p -subgroup Q of G with $H \leq Q$. Now let P be any Sylow p -subgroup of G . By Sylow's theorem, all Sylow p -subgroups are conjugate, so $P = gQg^{-1}$ for some $g \in G$. Since H is normal in G , we have $gHg^{-1} = H$, and therefore

$$H = gHg^{-1} \leq gQg^{-1} = P.$$

Since P was an arbitrary Sylow p -subgroup, H is contained in every Sylow p -subgroup of G .

5. (5 points) Prove that there are no simple groups of order 40.

Solution: Let G be a group of order $40 = 2^3 \cdot 5$. By Sylow's theorem, the number n_5 of Sylow 5-subgroups satisfies $n_5 \equiv 1 \pmod{5}$ and $n_5 \mid 8$. The divisors of 8 are 1, 2, 4, 8, and among these only $n_5 = 1$ satisfies $n_5 \equiv 1 \pmod{5}$.

Therefore there is a unique Sylow 5-subgroup P of G . Since P is the unique Sylow 5-subgroup, it must be normal in G (as gPg^{-1} is also a Sylow 5-subgroup for any $g \in G$, so $gPg^{-1} = P$). Since P is a proper nontrivial normal subgroup ($|P| = 5$, and $1 < 5 < 40$), G is not simple.

6. (5 points) Prove that every finite group has a composition series.

Solution: We proceed by strong induction on $|G|$.

Base case: If $|G| = 1$, then the trivial series $\{e\}$ is a composition series (it has no factors to check).

Inductive step: Assume every group of order less than $|G|$ has a composition series. If G is simple, then $\{e\} \trianglelefteq G$ is a composition series (the only factor is $G/\{e\} \cong G$, which is simple). If G is not simple, then G has a proper nontrivial normal subgroup. Among all proper nontrivial normal subgroups of G , choose a maximal one, say N (this exists since G is finite). Then G/N is simple: if G/N had a proper nontrivial normal subgroup $\bar{M}$, by the correspondence theorem $\bar{M} = M/N$ for some normal subgroup M of G with $N \subsetneq M \subsetneq G$, contradicting the maximality of N .

Since $|N| < |G|$, by the induction hypothesis N has a composition series

$$\{e\} = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_k = N.$$

Appending G at the top gives the series

$$\{e\} = N_0 \trianglelefteq N_1 \trianglelefteq \cdots \trianglelefteq N_k = N \trianglelefteq G,$$

where each factor N_{i+1}/N_i is simple (by the composition series of N) and G/N is simple by construction. This is a composition series for G .

7. (5 points) Prove that in a non-trivial finite p -group G , the center $Z(G)$ is non-trivial.

Solution: Let G be a finite p -group with $|G| = p^n$ for some $n \geq 1$. Consider the action of G on itself by conjugation, i.e., $g \cdot x = gxg^{-1}$. The orbits of this action are the conjugacy classes of G , and the class equation gives

$$|G| = |Z(G)| + \sum_i |G : G_{x_i}|,$$

where the sum runs over one representative x_i from each conjugacy class of size greater than 1, and G_{x_i} is the centralizer of x_i .

For each non-central element x_i , the orbit size $|G : G_{x_i}|$ divides $|G| = p^n$ and is greater than 1, so $p \mid |G : G_{x_i}|$. Since $p \mid |G|$ and p divides each term in the sum, it follows that $p \mid |Z(G)|$. In particular, $|Z(G)| \geq p > 1$, so $Z(G)$ is non-trivial.

8. (5 points) Let G be an abelian group of order m . If n divides m , prove that G has a subgroup of order n .

Solution: By the Fundamental Theorem of Finitely Generated Abelian Groups, we may write

$$G \cong \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_k^{a_k}},$$

where $m = p_1^{a_1} p_2^{a_2} \cdots p_k^{a_k}$ (not necessarily distinct primes). Since $n \mid m$, we can write $n = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k}$ where $0 \leq b_i \leq a_i$ for each i .

In each cyclic factor $\mathbb{Z}_{p_i^{a_i}}$, there is a unique subgroup of order $p_i^{b_i}$, namely the subgroup generated by $p_i^{a_i-b_i}$ (since cyclic groups have exactly one subgroup of each order dividing the group order). Call this subgroup $H_i \leq \mathbb{Z}_{p_i^{a_i}}$.

Then

$$H = H_1 \times H_2 \times \cdots \times H_k \leq \mathbb{Z}_{p_1^{a_1}} \times \mathbb{Z}_{p_2^{a_2}} \times \cdots \times \mathbb{Z}_{p_k^{a_k}} \cong G$$

is a subgroup of order $|H| = p_1^{b_1} p_2^{b_2} \cdots p_k^{b_k} = n$.